

# final Proj

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## R Markdown

```
# Load libraries
library(readxl)
library(caret)
```

```
## Loading required package: ggplot2
```

```
## Loading required package: lattice
```

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##   filter, lag
```

```
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(ggplot2)
library(pROC)
```

```
## Type 'citation("pROC")' for a citation.
```

```
##
## Attaching package: 'pROC'
```

```
## The following objects are masked from 'package:stats':
##
##   cov, smooth, var
```

```

# Load dataset
sq_data <- read_excel("C:/Users/prana/Desktop/UTA/5303-Stats and Computing/Final_project/Fertility_Dataset.xlsx")
sq_data_vis <- sq_data

# Clean and reformat column names
colnames(sq_data) <- gsub("\t", "", colnames(sq_data))

# Drop rows with missing values
sq_data <- na.omit(sq_data)

##### visualisation #####
#Comparison of Alcohol and Smoking Impacts on Sperm Quality

# Prepare data for smoking and alcohol impacts separately

alcohol_summary <- sq_data_vis %>%

  group_by(alcohol) %>%

  count(Diagnosis) %>%

  mutate(Proportion = n / sum(n))

smoking_summary <- sq_data_vis %>%

  group_by(smoking) %>%

  count(Diagnosis) %>%

  mutate(Proportion = n / sum(n))

# Plot side-by-side bar charts for comparison

p1 <- ggplot(alcohol_summary, aes(x = factor(alcohol), y = Proportion, fill = factor(Diagnosis))) +

  geom_bar(stat = "identity", position = "fill") +

  labs(title = "Impact of Alcohol Consumption on Sperm Quality", x = "Alcohol Consumption (1 = Every day, 4 = Hardly ever or never)", y = "Proportion", fill = "Diagnosis (1 = Normal, 0 = Altered)") +

  theme_minimal() +

  scale_fill_viridis_d()

p2 <- ggplot(smoking_summary, aes(x = factor(smoking), y = Proportion, fill = factor(Diagnosis))) +

  geom_bar(stat = "identity", position = "fill") +

```

```

labs(title = "Impact of Smoking Habits on Sperm Quality", x = "Smoking Habits (-1=Never, 0=Occasional, 1=Daily)", y = "Proportion", fill = "Diagnosis (1 = Normal, 0 = Altered)") +

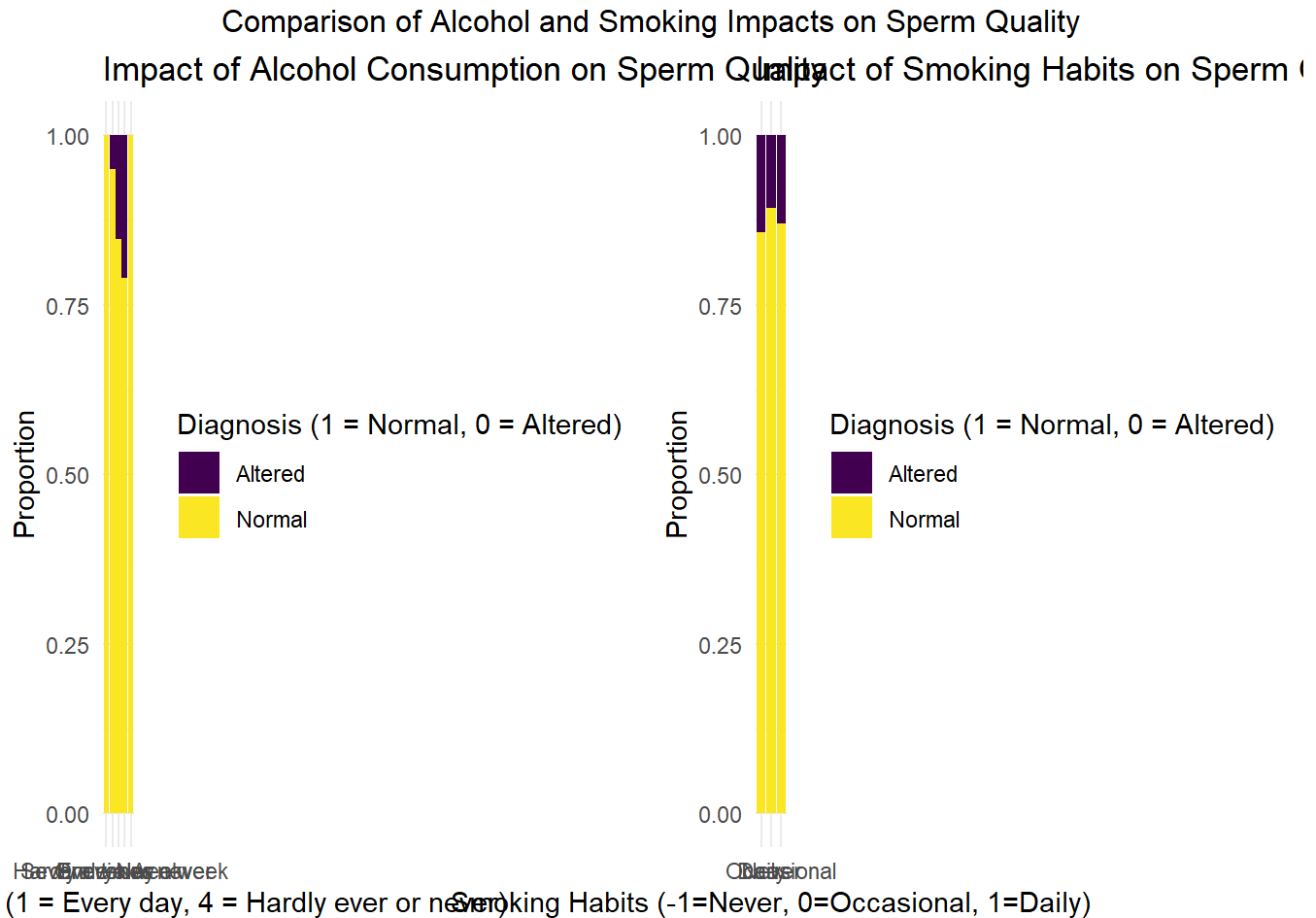
theme_minimal() +

scale_fill_viridis_d()

# Combine the plots

gridExtra::grid.arrange(p1, p2, ncol = 2, top = "Comparison of Alcohol and Smoking Impacts on Sperm Quality")

```



```

#####
# 7. Surgical Interventions vs. Diagnosis (Stacked Bar Chart)
surgical_summary <- sq_data_vis %>%
  group_by(surgical_intervention, Diagnosis) %>%
  summarise(Count = n()) %>%
  ungroup()

```

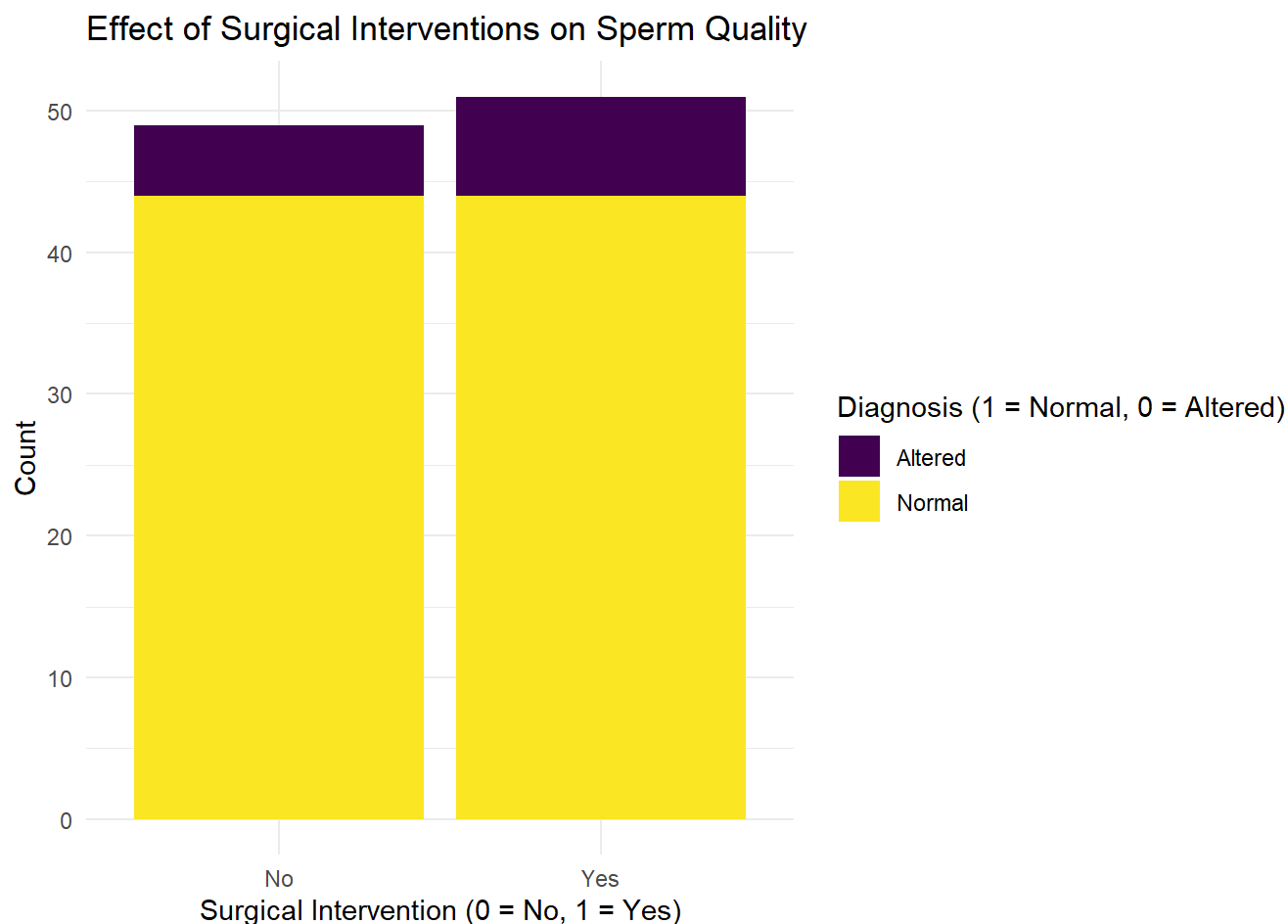
```

## `summarise()` has grouped output by 'surgical_intervention'. You can override
## using the `.groups` argument.

```

```
surgical_plot <- ggplot(surgical_summary, aes(x = factor(surgical_intervention), y = Count, fill
= factor(Diagnosis))) +
  geom_bar(stat = "identity", position = "stack") +
  labs(title = "Effect of Surgical Interventions on Sperm Quality", x = "Surgical Intervention
(0 = No, 1 = Yes)", y = "Count", fill = "Diagnosis (1 = Normal, 0 = Altered)") +
  theme_minimal() +
  scale_fill_viridis_d()

print(surgical_plot)
```



#### # 8. High Fevers vs. Diagnosis (Diverging Bar Chart)

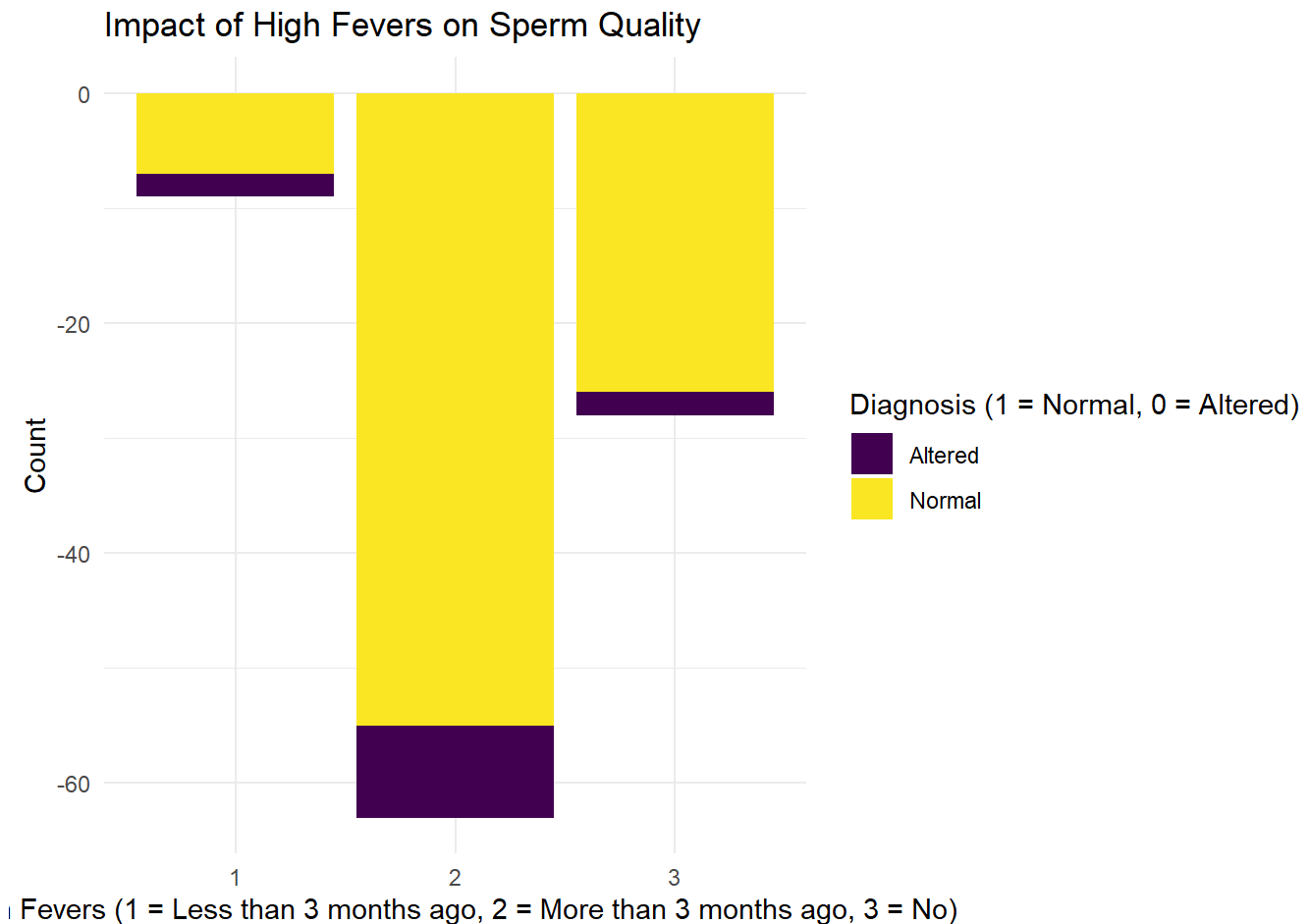
```
sq_data_vis$high_fevers <- recode(sq_data_vis$high_fevers, "Less than 3 months ago" = 1, "More t
han 3 months ago" = 2, "No" = 3)
```

```
fever_summary <- sq_data_vis %>%
  group_by(high_fevers, Diagnosis) %>%
  summarise(Count = n()) %>%
  mutate(Divergence = ifelse(Diagnosis == 1, Count, -Count))
```

```
## `summarise()` has grouped output by 'high_fevers'. You can override using the
## `.groups` argument.
```

```
fever_plot <- ggplot(fever_summary, aes(x = factor(high_fevers), y = Divergence, fill = factor(Diagnosis))) +
  geom_bar(stat = "identity") +
  labs(title = "Impact of High Fevers on Sperm Quality", x = "High Fevers (1 = Less than 3 months ago, 2 = More than 3 months ago, 3 = No)", y = "Count", fill = "Diagnosis (1 = Normal, 0 = Altered)") +
  theme_minimal() +
  scale_fill_viridis_d()

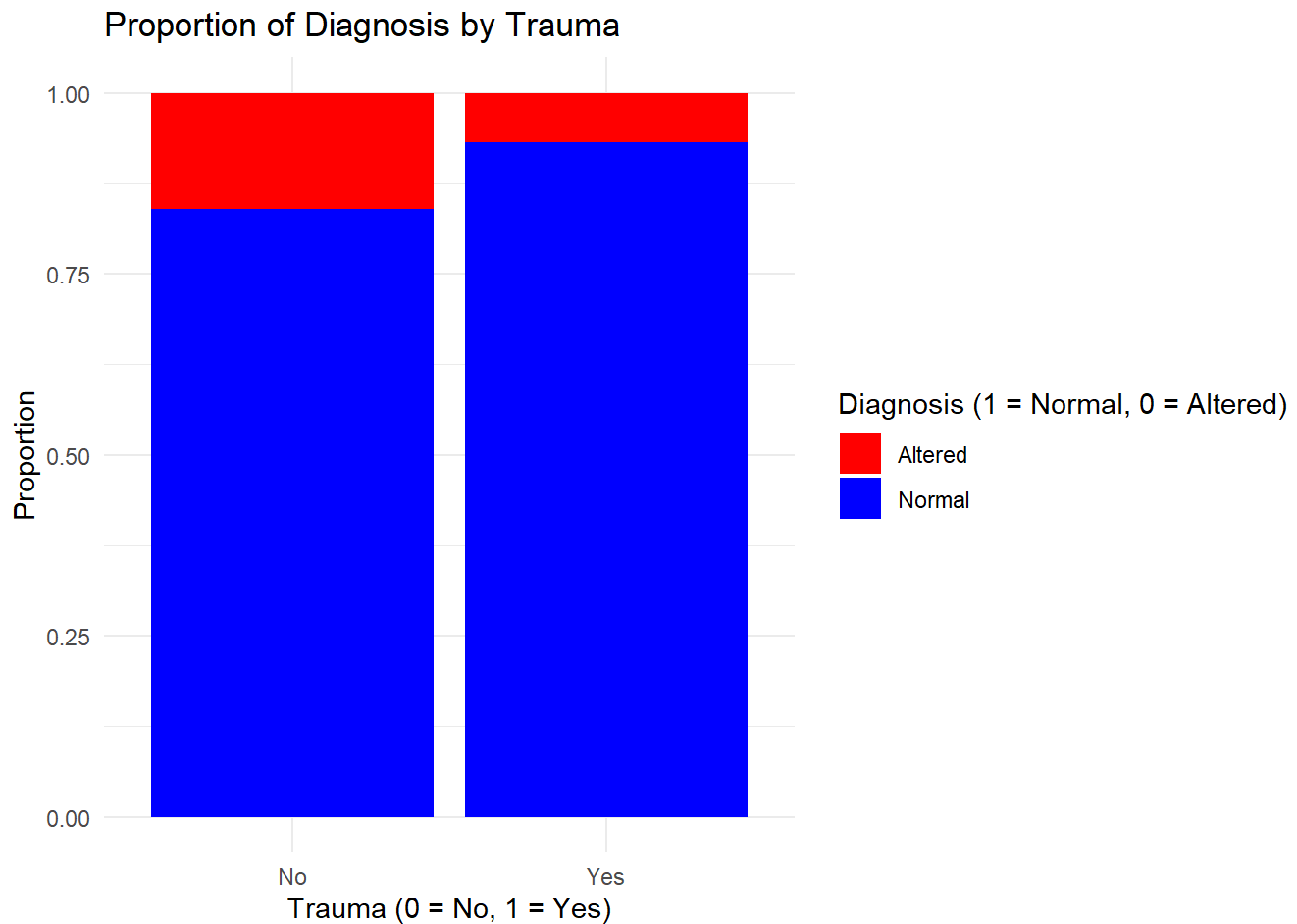
print(fever_plot)
```



```
#####
trauma_summary <- sq_data_vis %>%
  group_by(accident, Diagnosis) %>%
  summarise(Count = n()) %>%
  group_by(accident) %>%
  mutate(Proportion = Count / sum(Count)) %>%
  ungroup()
```

```
## `summarise()` has grouped output by 'accident'. You can override using the
## `.groups` argument.
```

```
trauma_plot <- ggplot(trauma_summary, aes(x = factor(accident), y = Proportion, fill = factor(Diagnosis))) +  
  geom_bar(stat = "identity", position = "stack") +  
  labs(title = "Proportion of Diagnosis by Trauma", x = "Trauma (0 = No, 1 = Yes)", y = "Proportion", fill = "Diagnosis (1 = Normal, 0 = Altered)") +  
  theme_minimal() +  
  scale_fill_manual(values = c("red", "blue"))  
  
print(trauma_plot)
```



# ##### Data Preprocessing #####

## # Recode categorical variables

```
sq_data$Season <- recode(sq_data$Season, "Winter" = -1, "Spring" = -0.33, "Summer" = 0.33, "Fall" = 1)
sq_data$Child_Diseases <- recode(sq_data$Child_Diseases, "Yes" = 0, "No" = 1)
sq_data$accident <- recode(sq_data$accident, "Yes" = 0, "No" = 1)
sq_data$surgical_intervention <- recode(sq_data$surgical_intervention, "Yes" = 0, "No" = 1)
sq_data$high_fevers <- recode(sq_data$high_fevers, "Less than 3 months ago" = -1,
                             "More than 3 months ago" = 0, "No" = 1)
sq_data$alcohol <- recode(sq_data$alcohol, "Every day" = 0.2, "Several times a week" = 0.6,
                          "Once a week" = 0.8, "Hardly ever or never" = 1)
sq_data$smoking <- recode(sq_data$smoking, "Never" = -1, "Occasional" = 0, "Daily" = 1)
sq_data$Diagnosis <- recode(sq_data$Diagnosis, "Normal" = 1, "Altered" = 0)
```

## # Normalize continuous variables

```
sq_data$Age <- scales::rescale(sq_data$Age, to = c(0, 1))
sq_data$hrs_sitting <- scales::rescale(sq_data$hrs_sitting, to = c(0, 1))
```

## # Balance classes using oversampling

```
minority <- sq_data[sq_data$Diagnosis == 0, ]
majority <- sq_data[sq_data$Diagnosis == 1, ]
oversampled_minority <- minority[sample(1:nrow(minority), nrow(majority), replace = TRUE), ]
balanced_data <- rbind(majority, oversampled_minority)
```

## # Split data into training and testing sets

```
set.seed(42)
trainIndex <- createDataPartition(balanced_data$Diagnosis, p = 0.8, list = FALSE)
trainData <- balanced_data[trainIndex, ]
testData <- balanced_data[-trainIndex, ]
```

## # Fit logistic regression model

```
logit_model <- glm(Diagnosis ~ ., data = trainData, family = binomial())
```

## # Predict on test data

```
logit_prob <- predict(logit_model, newdata = testData, type = "response")
logit_pred <- ifelse(logit_prob > 0.5, 1, 0)
```

## # Confusion matrix and accuracy

```
conf_matrix <- table(Predicted = logit_pred, Actual = testData$Diagnosis)
print(conf_matrix)
```

```
##           Actual
## Predicted  0   1
##           0 14   4
##           1   3 13
```

```
accuracy <- sum(diag(conf_matrix)) / sum(conf_matrix)
print(paste("Accuracy:", round(accuracy, 2)))
```

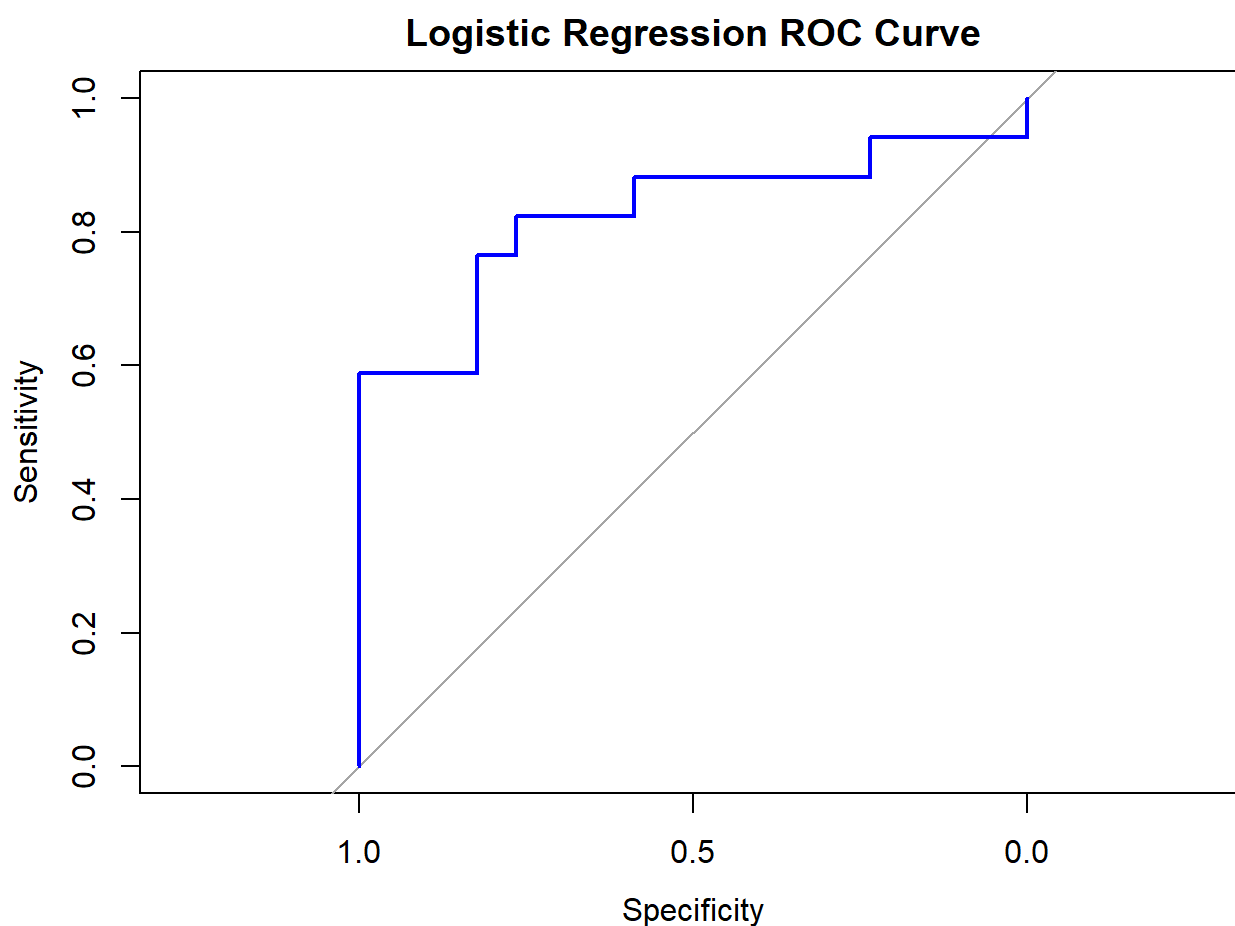
```
## [1] "Accuracy: 0.79"
```

```
# ROC curve and AUC
roc_obj <- roc(testData$Diagnosis, logit_prob)
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
plot(roc_obj, main = "Logistic Regression ROC Curve", col = "blue")
```



```
auc_value <- auc(roc_obj)
print(paste("AUC:", round(auc_value, 2)))
```

```
## [1] "AUC: 0.83"
```